👩‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 2 · STANDARD 6.NOS.3

Multiply Decimals

Lesson 2-12 · Teacher Copy

Name: _____ **Date:** _____ **Class:** _____

My Notes

Level 3 Enrichment

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can multiply decimals and place the decimal point correctly in the product.

Language Objective: I can explain my work using the words product, decimal point, estimate, and decimal places.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Product | **Decimal point** | **Estimate** | **Place value** | **Decimal places**

Level 3: try the blanks from memory first, then check the bank.

1 The answer to a multiplication problem is the _____ .

2 The dot that separates whole numbers from parts of a whole is the _____ .

3. Finding an answer that is close but not exact, to check if a product is reasonable, is to _____.

4. The value a digit has based on its position is its _____.

5. To place the point in a product, I count the total number of _____ in both factors.

Write About the Math

How do you multiply decimals to find the total cost? How can estimation help you check?
¿Cómo multiplicas decimales para hallar el costo total? ¿Cómo te puede ayudar estimar a comprobarlo?

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **Product** — *Producto*
- ☐ **Decimal point** — *Punto decimal*
- ☐ **Estimate** — *Estimar*
- ☐ **Place value** — *Valor posicional*
- ☐ **Decimal places** — *Cifras decimales*

Model: *Rounding the factors gives a reasonable estimate of the product.* — Students round to about 5 × $13 = $65 (or 4 × $13) to predict the cost is in the $50-$65 range, naming 4.5 and 12.60 as the factors.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The total cost is $___ because 3.5 × $6.80 = $___, and I can estimate by rounding to ___ × ___ = ___.**
- **My answer is ___ because ___.**
  Mi respuesta es ___ porque ___.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ___.**
  Mi afirmación es ___.
- **My evidence is ___, so ___.**
  Mi evidencia es ___, entonces ___.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Product
2. **Notes 2:** Decimal point
3. **Notes 3:** Estimate
4. **Notes 4:** Place value
5. **Notes 5:** Decimal places
6. **Watch for this mistake:** *A common mistake in Multiply Decimals is counting decimal places from only one factor instead of adding the decimal places from BOTH factors. For example, in 0.4 × 0.06, a student multiplies 4 × 6 = 24, then places the decimal point using*

just one factor's place count (1 place), writing 0.24 instead of the correct answer: 0.4 has 1 decimal place and 0.06 has 2 decimal places, for a total of 3, so 24 becomes 0.024. Before you submit, count the decimal places in EACH factor separately and add them together — that total is how many decimal places belong in your final answer.

7. **Try It 1:** B. $0.24 — *6 × 4 = 24. There are 2 total decimal places (1 + 1), so 0.6 × 0.4 = 0.24, which is $0.24.*
8. **Try It 2:** B. $3.00 — *25 × 12 = 300. There are 2 total decimal places (1 + 1), so 2.5 × 1.2 = 3.00 = $3.00.*